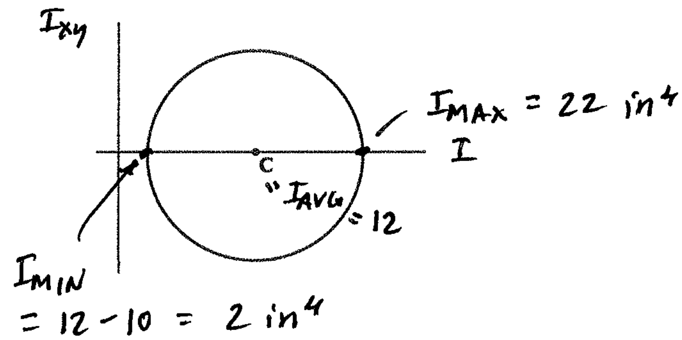
Simple Mohr's Circle Computation Questions Typical for Exam 4

1. Given I_{AVG} and radius (or diameter), determine I_{MAX} and/or I_{MIN}.
(The problem may say that the "center" is at some value. Realize that the "center" is I_{AVG}.

Question 1
Given I_{AVG} and radius, find I_{MAX} and I_{MIN}.

Center I_{AVG} =	12.00	in^4
Radius R =	10.00	in^4

Picture (usually not provided on the exam, but provided here to help you visualize the solution)



Solution:

$$I_{AVG} = \text{center of MC} = \frac{I_x + I_y}{2}$$

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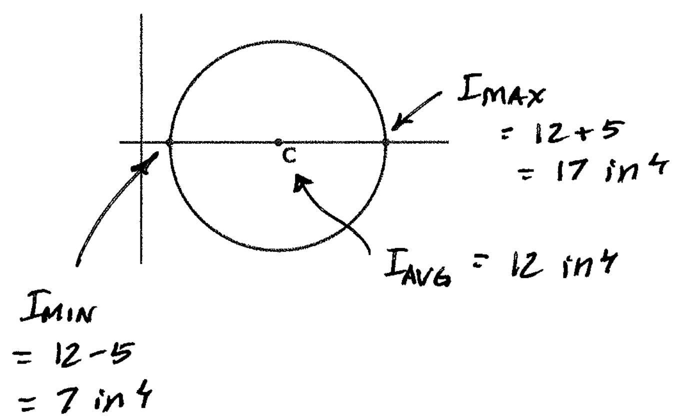
2. Given I_{AVG} and diameter, determine I_{MAX} and/or I_{MIN}.
(The problem may say that the "center" is at some value. Realize that the "center" is I_{AVG}.

Question 2:
Given I_{AVG} and diameter, find I_{MAX} and I_{MIN}.

Center I_{AVG} =	12.00	in^4
Diameter d =	10.00	in^4

Picture (usually not provided on the exam, but provided here to help you visualize the solution)



Solution:

$$dia = 10 \ in^4$$

$$R = \frac{dia}{2} = 5 \ in^4$$

3. Given $\mathbf{I}_{AVG}$ and point B ($\mathbf{I}_B$, $\mathbf{I}_{BD}$) (note that point B is sometimes called a "reference point on the circle), determine the following:

Radius (R), $\mathbf{I}_{MAX}$, $\mathbf{I}_{MIN}$, point D (90 degrees from point B on the area), Angles to principal axes θ_{p1}, θ_{p2}.

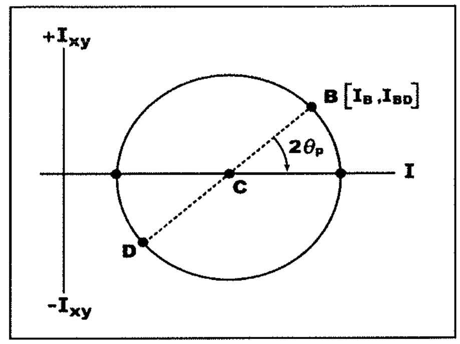
(The problem may say that the "center" is at some value. Realize that the "center" is $\mathbf{I}_{AVG}$.

Question 3:

Given I_{AVG} and point B [I_B,I_{BD}]:

	Center I_{AVG} =	12.00	in^4
Coordinates of point B [I_B,I_{BD}]:	**I_B =**	19.88	in^4
	I_{BD} =	6.16	in^4

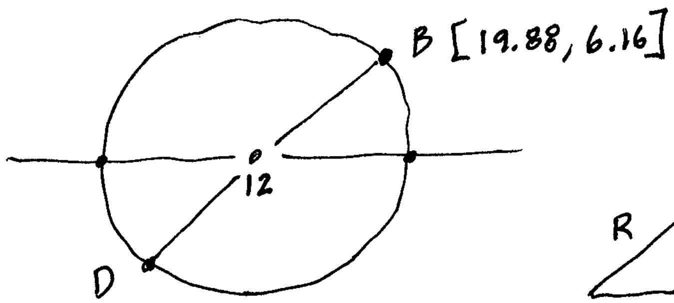
Picture (usually not provided on the exam, but provided here to help you visualize the solution)

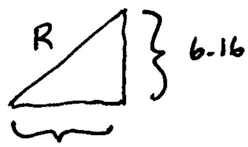


Please determine:

Radius R, I_{MAX}, I_{MIN}, point D [I_D,I_{BD}] (90 deg from point B on area), θ_{p1}, θ_{p2}

Solution:





$19.88 - 12 = 7.88$

$R = \sqrt{7.88^2 + 6.16^2}$

$= \text{norm}([7.88, 6.16])$

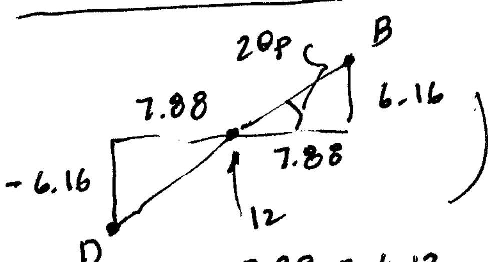
$\boxed{R = 10.0 \text{ in}^4}$

Since $R = 10$ in^4,

$$\boxed{\begin{aligned} I_{MAX} &= 12 + 10 = 22 \text{ in}^4 \\ I_{MIN} &= 12 - 10 = 2 \text{ in}^4 \end{aligned}}$$

Coords of D = Mirror Image of B :

$\boxed{\text{Coords of D} = [4.12, -6.16] \text{ in}^4}$



$12 - 7.88 = 4.12$

$$2\theta_p = \tan^{-1} \frac{6.16}{7.88}$$

$$= 38.0°$$

$$\boxed{\begin{aligned} \theta_{p_1} &= 19.0° \\ \theta_{p_2} &= 19 + 90 = 109° \end{aligned}}$$

4. Given I_{AVG} , Radius (R), and angle ϕ CCW from I_{MAX}, determine the following: I_{MAX}, I_{MIN}, and point B.

Question 4

Given I_{AVG} , Radius and angle ϕ CCW from I_{MAX}:

	Center I_{AVG} =	12.00	in^4	
	Radius R =	10.00	in^4	
(Point B in upper right quadrant of MC)	ϕ =	15.0	degrees	(= 2θ on MC)

Please determine: **I_{MAX}, I_{MIN}, point B [I_B,I_{BD}] (at ϕ deg CCW from I_{MAX})**

Solution: $I_{AVG} = 12 \ in^4$

$R = 10$

I_{AVG}

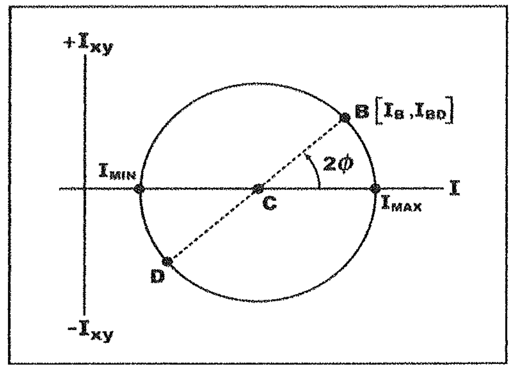
$30°$

$R \sin 30 = 5$

$R \cos 30 = 8.66$

$$\boxed{B = [18.66, 5] \ in^4}$$

Picture (usually not provided on the exam, but provided here to help you visualize the solution)



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5. Given I_{AVG} , Radius (R), and angle ϕ CW from I_{MIN}, determine the following: I_{MAX}, I_{MIN}, and point B.

Question 5:

Given I_{AVG} , Radius and angle ϕ CW from I_{MIN}:

	Center I_{AVG} =	12.00	in^4	
	Radius R =	10.00	in^4	
(Point B in upper left quadrant of MC)	ϕ =	15.0	degrees	(= 2θ on MC)

Please determine: **I_{MAX}, I_{MIN}, point B [I_B,I_{BD}] (at ϕ deg CW from I_{MIN})**

Solution:

B

10

$30°$

5

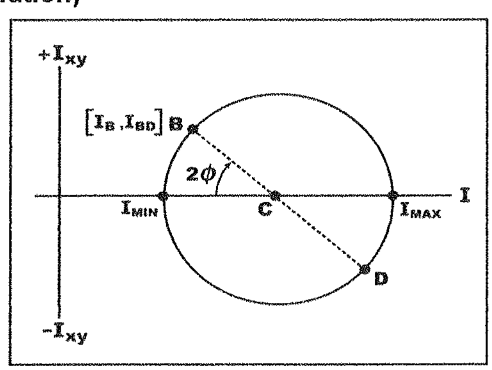
8.66

$I_{AVG} = 12$

$$\boxed{B = [3.34, 5] \ in^4}$$

$3.34 = 12 - 8.66$

Picture (usually not provided on the exam, but provided here to help you visualize the solution)



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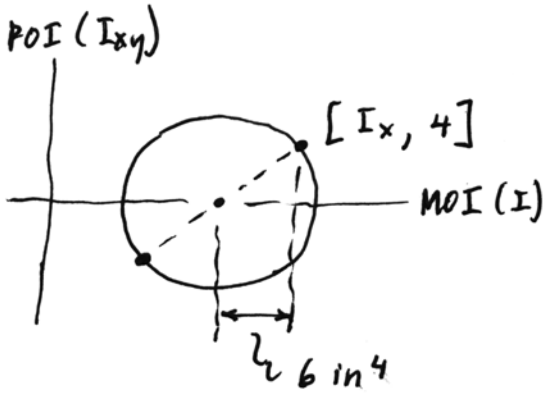
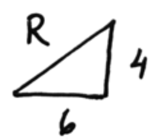
6. Given: I at a particular point is 6 in⁴ larger or smaller than the center. POI at that point is 4 in⁴. Find R





$$R = \sqrt{6^2 + 4^2}$$

$$\boxed{R = 7.21 \text{ in}^4}$$

7. Example Problem

Two axes P + Q.

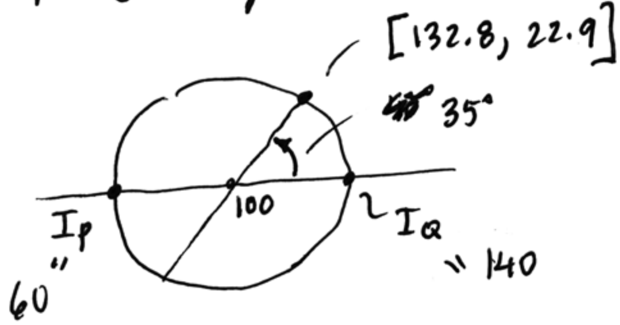
$I_P = 60 \text{ in}^4$

$I_Q = 140 \text{ in}^4$

Area is symmetric about axis P.

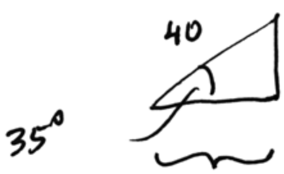
This tells us that $I_{xy} = 0$.

Thus I_P + I_Q are principals.



Find I_{xy} on MC WRT axes rotated at 35 ~~40~~° on MC. CCW

$$R = \frac{140 - 60}{2} = 40$$



$\} \; 40 \sin 35 = 22.9$

$40 \cos 35 = 32.8$